Object state between contacts

Second-week research report

What does current shape reveal, and what does interaction history add?

Spatial 2D contact with fast and slow internal relaxation

The current shape carries substantial information.
Contact and recovery history reduce the remaining prediction error.

28 October 2026

What changed since the first report

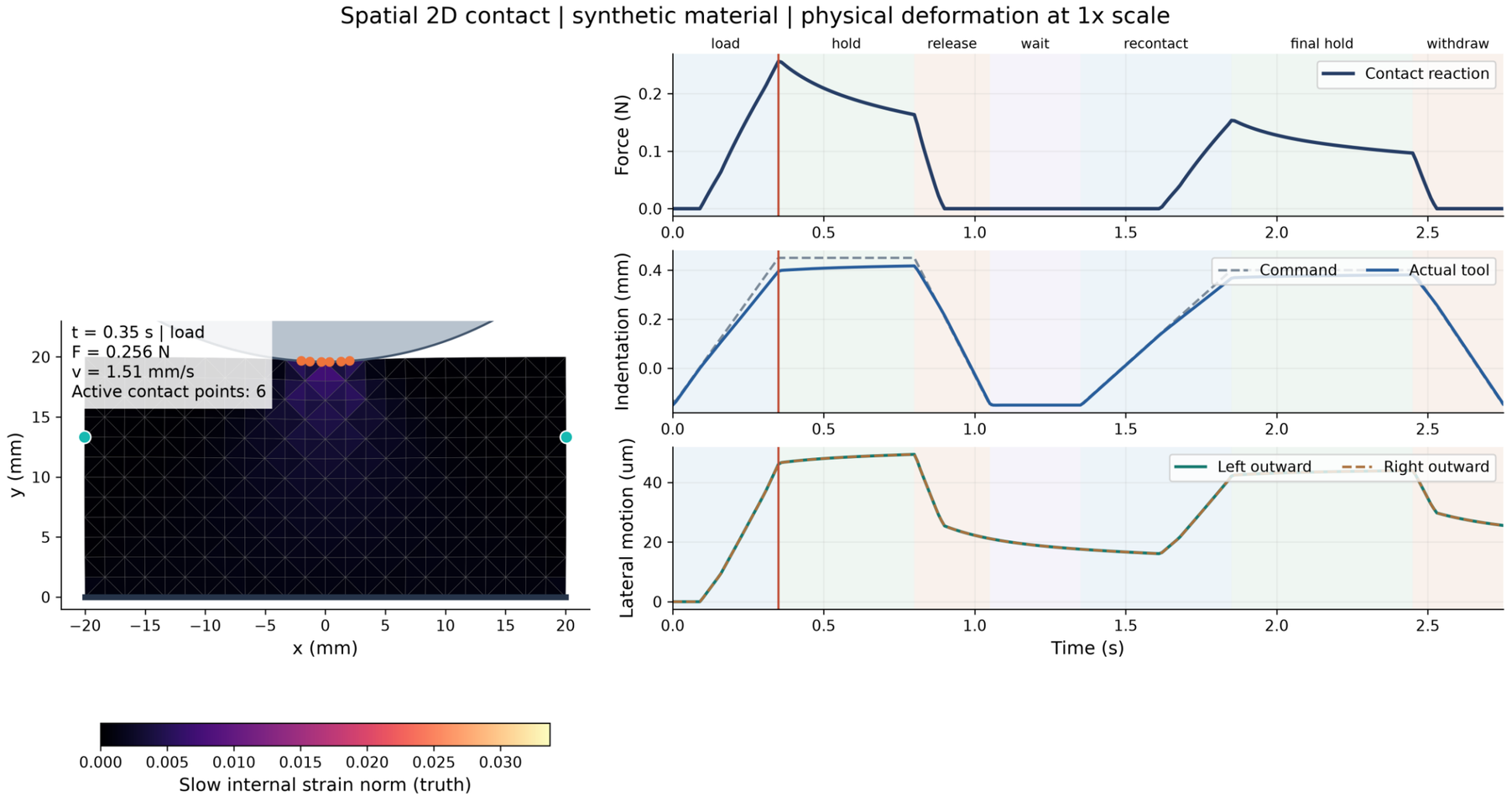
First report	This week
1D motion with two internal variables	Spatial 2D FEM with lateral deformation
History predicts different future responses	Separate current shape from additional history
A short classical observer can already work well	Separate waiting time from informative contact data
Shared feedback gave little extra action benefit	Keep action benefit as a separate, unfinished test

The meeting feedback motivated the 2D contact and lateral-response test.

The 2D-to-3D and lateral-response feedback comes from the recorded meeting recap.

A spatial body produces the force and lateral response

Fixed bottom, free sides, one round rigid indenter. The deformation comes from the FEM solution.



Recorded 2D pilot. Synthetic material. Quasistatic body with a dynamic tool. Deformation shown at 1x scale.

The physical state contains more than the current measurement

Physical state

Fast and slow internal strain fields
Tool position and velocity

Relaxation times: 0.12 s and 0.90 s
Equilibrium determines deformation and contact.

Available observation

Tool position, velocity and total force
Optional left/right lateral displacement

During detached waiting, force is zero.
Lateral shape still changes during recovery.

Classical state estimate

Fit six nonnegative coefficients for the two internal fields.
Propagate the estimate with the same FEM and a future command.

The estimate is a restricted physical representation. Current shape alone is not a sufficiency proof.

The comparison changes only the available information

Same physical histories, prediction time, future solver and observed noise draw.

Information	Motion + force	Motion + force + lateral shape
Current sample	Position, velocity, total force	The same sample plus two lateral markers
Recent history	The same channels over 0.2 or 0.8 s	The same window plus lateral history

24 independent loading histories

Three noise draws per history.
History, rather than noise draw, is the statistical unit.

Three separate calibration histories

Equal tuning budgets select one shared preset.
Estimate once, then predict Slow and Fast press.

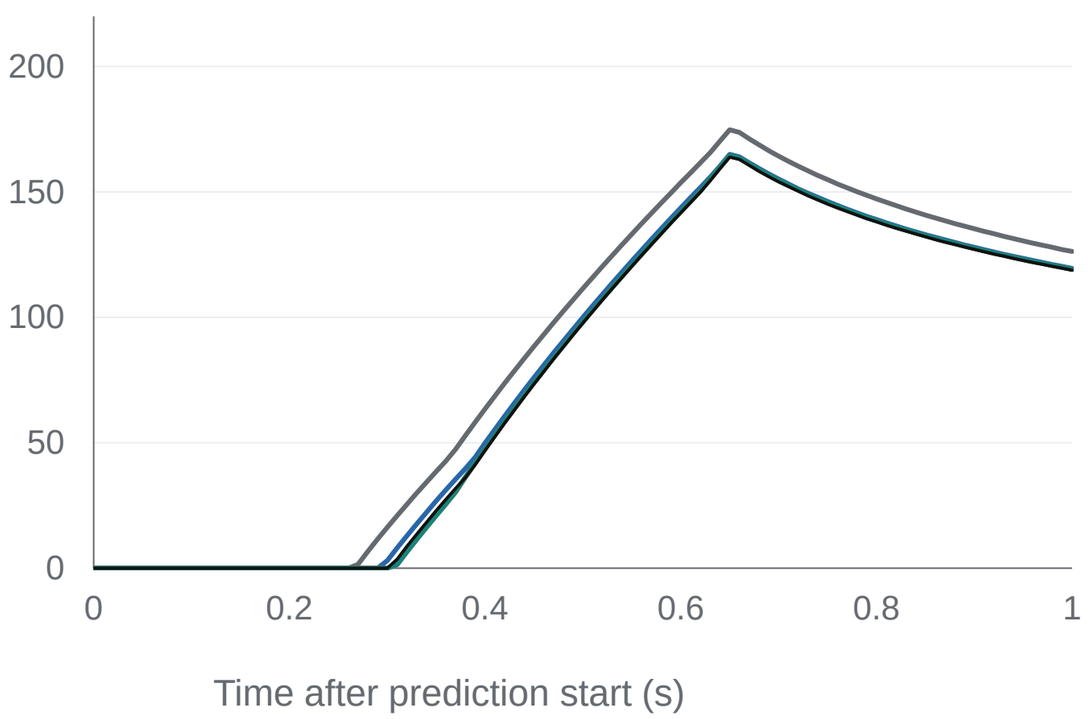
Prediction starts at 1.40 s. Disabled channels are absent from both initialization and fitting.

The same future command exposes different prediction errors

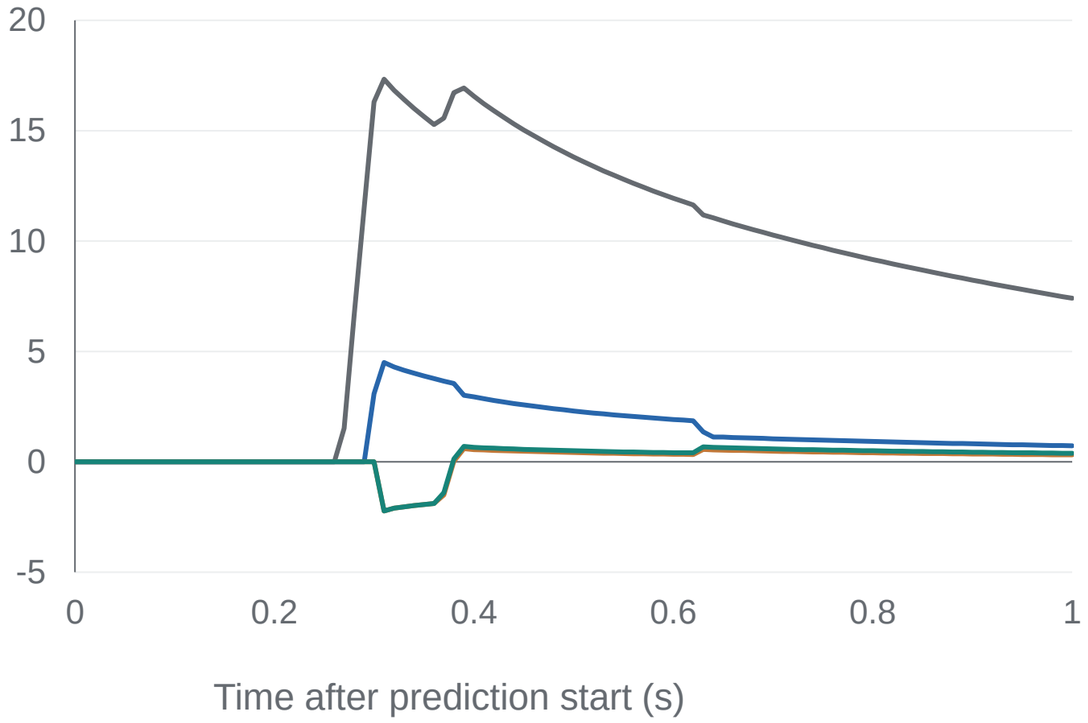
First prespecified test history, noise draw 0, Slow press. History conditions use the recent 0.8 s.

Current motion + force Current + lateral shape 0.8 s motion + force 0.8 s + lateral shape True response

Contact force (mN)



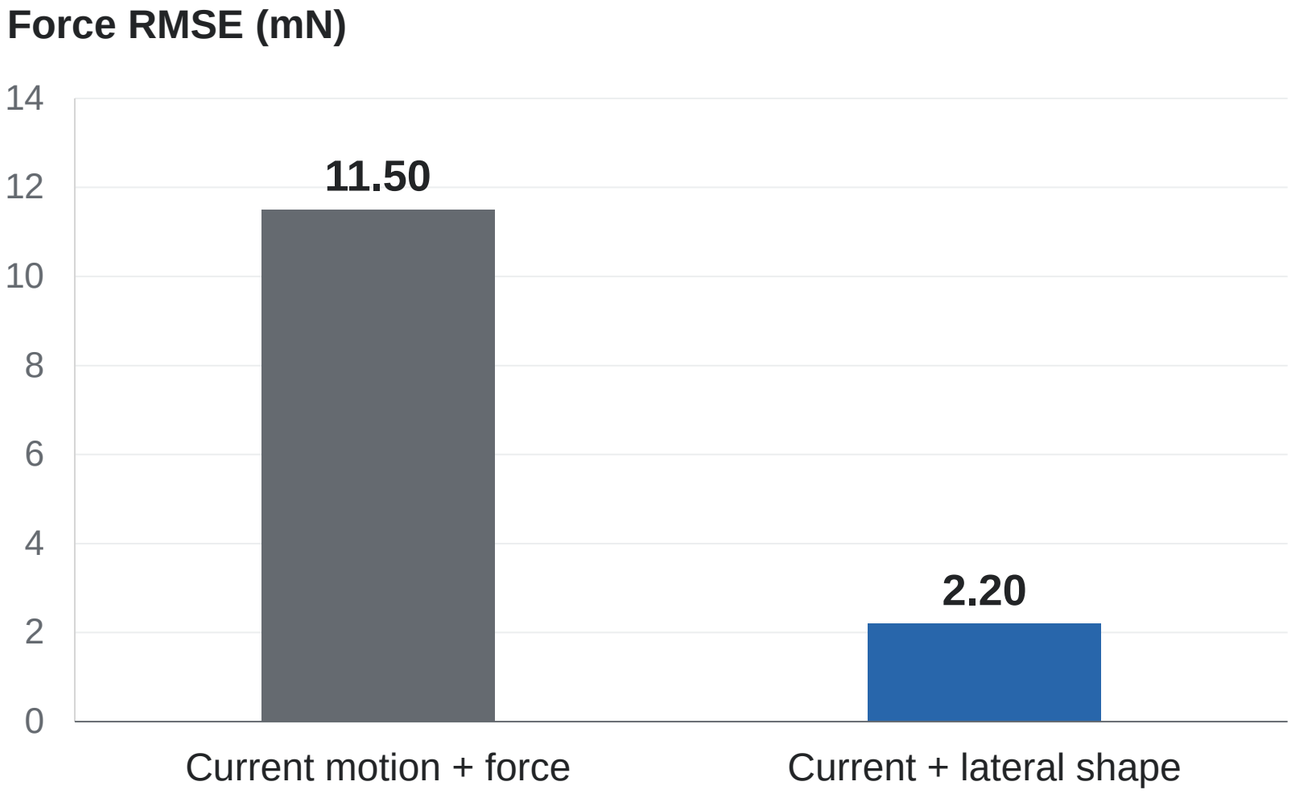
Prediction error (mN)



All predictions start from the same observed tool state. Future achieved motion is not supplied to the estimator.

Current lateral shape carries substantial predictive information

Force RMSE for Slow press, averaged within history and then across 24 histories.



9.31 mN

paired reduction in error

95% interval: 7.06 to 11.57 mN

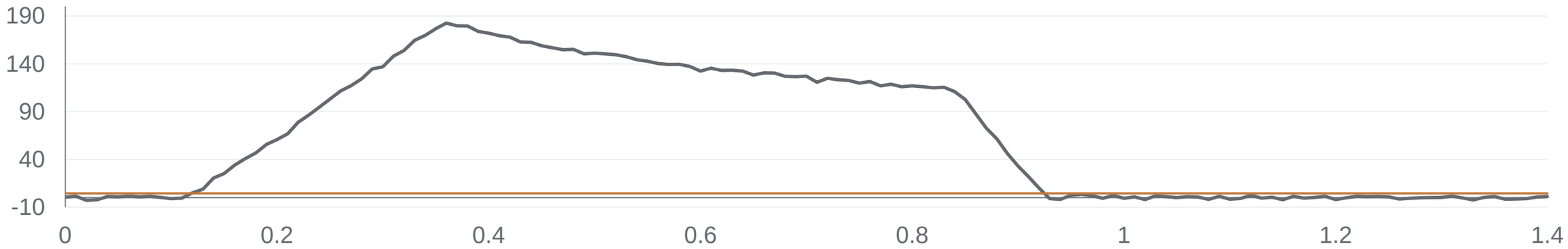
Shape exposes recovery while contact force is zero.

Both lateral markers share one underlying mode in this symmetric model. Their measurement noise is independent.

The recent 0.2 s contains waiting, while earlier data contain contact

Example history: load to 0.36 s, hold to 0.84 s, release to 1.10 s, wait to 1.40 s.

Observed contact force (mN)



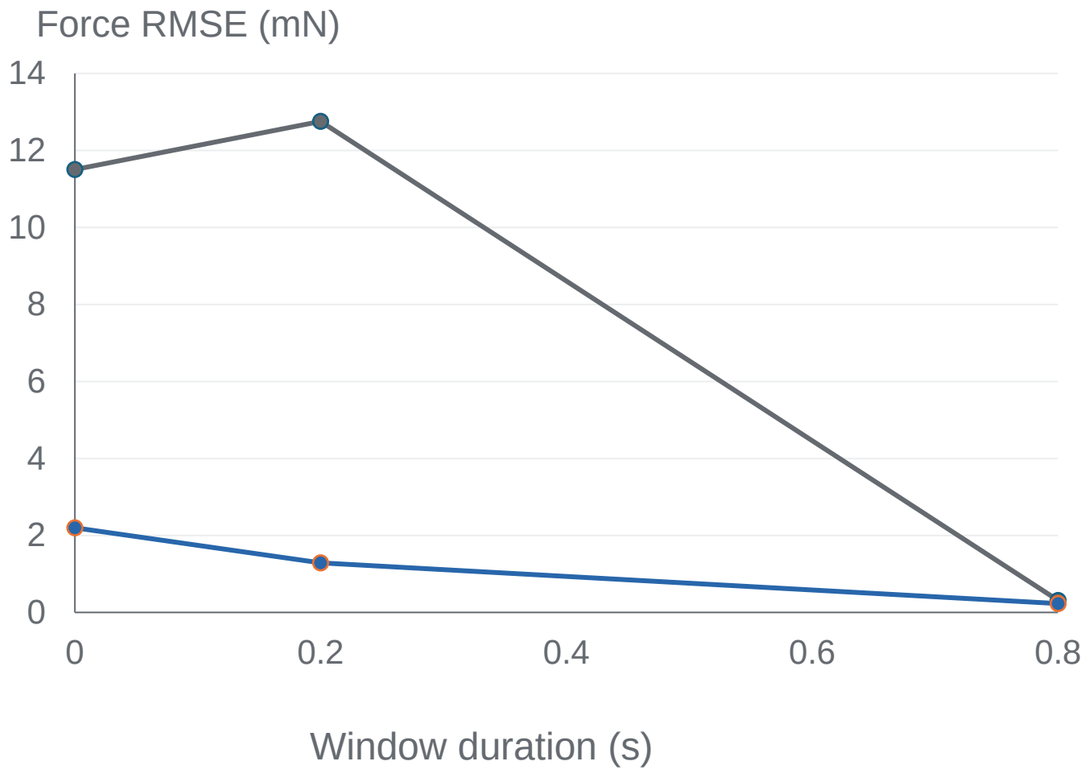
Estimator input	Observation update	Propagation	Contact duration
Recent wait	1.20-1.40 s	None	0 s
Recent 0.8 s	0.60-1.40 s	None	0.32 s
Earlier contact	0.72-0.92 s	0.92-1.40 s	0.20 s

Contact selection uses observed force > 4.5 mN and scans prior force. The contact method spans 0.68 s of motion here.

History helps when it includes informative observations

Mean Slow-press force RMSE. The nominal-noise result uses all 24 independent histories.

Recent window



Without lateral **With lateral**

Earlier contact plus propagation

Channels	Recent wait	Earlier contact
Without lateral	12.75 mN	0.48 mN
With lateral	1.29 mN	0.39 mN

Current lateral: 2.20 mN
Recent lateral wait: 1.29 mN
Recent 0.8 s with lateral: 0.23 mN

Recovery dynamics and noise averaging both may help. Contact and duration are not completely disentangled.

The same estimate also predicts a faster press

No refitting on the future action. Slow press alone selects the calibration preset.

Information used	Slow press	Fast press
Current + lateral shape	2.20	2.52
Recent 0.2 s + lateral shape	1.29	1.26
Earlier contact + propagation, with lateral	0.39	0.42
Recent 0.8 s + lateral shape	0.23	0.21

The estimate transfers across these two future commands.

Units: mN force RMSE. This is prediction reuse. The 2D study has not yet tested improvement in closed-loop actions.

The classical observer is accurate, but repeated fitting is costly

Recorded result	What it changes
0.23 mN with the recent 0.8 s and lateral sensing	A neural model has no demonstrated accuracy gap to fill.
27.24 s median inference for that window	A recursive update should avoid refitting the same past.
201 of 576 fits reached the evaluation limit	Optimization status and update cost need separate checks.
All 72 force-only wait fits retained the initial coefficients	Waiting force cannot supply the missing state information.

The current estimate has no uncertainty and uses a restricted six-coefficient basis.

Timing is measured during an eight-worker CPU batch. Accurate simulation prediction does not imply real-time control.

Next: retain the contact estimate through release and recovery

Immediate experiment

Compare recursive classical updates with the current window fits.
Measure prediction error, update time and uncertainty at several prediction times.

A small control to explain the history gain

Compare evolving recovery observations with repeated measurements of a fixed state.
This separates dynamic information from noise averaging.

Then test actions under common feedback

First check whether true-state information can help the selected task.
Full 3D robot simulation follows the 2D decision test, before real robot operation.

For discussion: which manipulation needs this state before feedback catches up, and which observable should define success?

Appendix: all recorded prediction errors

Force RMSE in mN. Twenty-four histories, three independent nominal-noise draws per history.

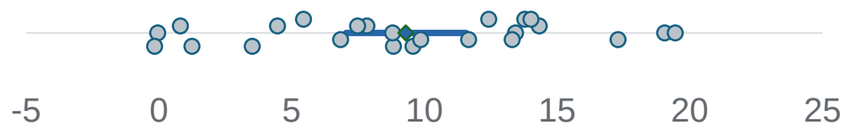
Information	Slow press	Fast press
Current motion + force	11.504	12.764
Current + lateral	2.198	2.516
Recent 0.2 s, without lateral	12.753	14.323
Recent 0.2 s, with lateral	1.286	1.256
Recent 0.8 s, without lateral	0.314	0.317
Recent 0.8 s, with lateral	0.230	0.207
Earlier contact + propagation, without lateral	0.476	0.514
Earlier contact + propagation, with lateral	0.389	0.415
Full replay from a known relaxed start [privileged]	0.0065	0.0075
True internal fields and the same solver [privileged]	0.000	0.000

The references supply privileged state or initialization. They are not absolute optimal-control bounds.

Appendix: paired gains across complete loading histories

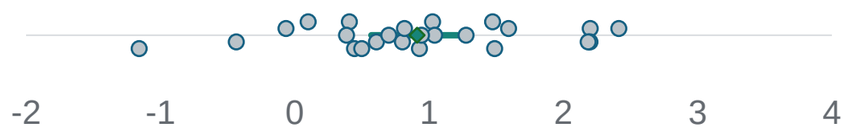
Positive values mean lower force prediction error for Slow press.

Add current lateral shape



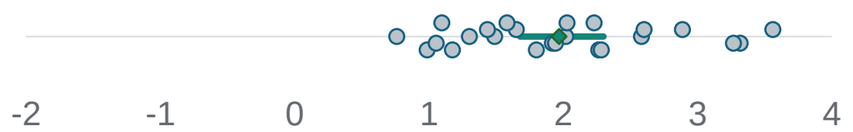
9.31 mN
[7.06, 11.57]

Add 0.2 s lateral wait



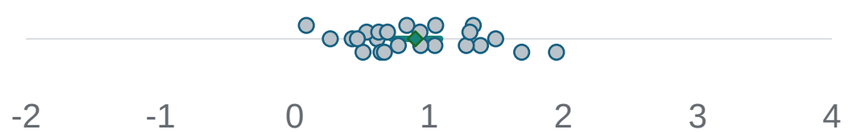
0.91 mN
[0.57, 1.27]

Add 0.8 s lateral history



1.97 mN
[1.68, 2.30]

Contact + propagation vs lateral wait



0.90 mN
[0.72, 1.08]

Three noise draws average within each history. 2,000 history bootstrap resamples. Pointwise exploratory 95% intervals.

Appendix: physical checks and reproducibility

Item	Recorded status
Material and geometry	40 x 20 mm body, 10 mm thickness, 576 triangles, dt = 0.01 s
Synthetic relaxation model	E = 20 / 40 / 60 kPa, Poisson ratio 0.35, relaxation 0.12 / 0.90 s
Nominal sensor noise	Position 0.5 um, velocity 0.01 mm/s, force 1.5 mN, lateral 1 um
Physics exceptions	Test penetration 5.030 um and calibration 5.015 um exceed the 5 um limit
Computation and consistency	0 computation exceptions, 10 regression tests and 7 data checks passed
Notebook and batch runtime	Fresh Docker kernel Run All: 7.61 s. Main batch: 965.85 s on 8 CPU workers

Both penetration overruns remain in the data. Real-material calibration is still missing.

Questions for discussion

Research focus

We want to study hidden state beyond the current observation.
Could too much focus on elasticity shift the project toward soft robotics?

Physical experiment

We can build the model. How should we carry out
these experiments on a real robot?

Broader research question

After studying hidden state through fast and slow forces,
how do we develop the broader idea from this specific case?